BITCOIN SUSTAINABILITY WRAPPED

2025 YEAR IN REVIEW

BRIDGING BITCOIN, CLEAN ENERGY, AND INSTITUTIONAL CAPITAL

SUSTAINABLE BITCOIN PROTOCOL

LETTER FROM THE CEO

2025 marked a clear inflection point for Sustainable Bitcoin Protocol (SBP) and for bitcoin mining's role within global energy systems and capital markets. For the first time, the defining questions facing the industry extended beyond price and hashrate toward energy sourcing, system integration, and capital durability.

SBP was founded on the premise that as institutional ownership of bitcoin expanded, energy behavior would no longer remain opaque. Markets would ultimately desire credible, verifiable differentiation between climate-aligned and undifferentiated bitcoin production, and that it's possible to bring the trillions of sidelined climate capital to bitcoin, one of the greatest energy technologies of our time. In 2025, that shift began to materialize.

In 2025, SBP scaled to include miners representing **nearly 25% of global bitcoin hashrate**, spanning **five continents and more than two dozen countries**. Participants included **public, private, and sovereign bitcoin miners**, reflecting broad-based adoption across operating models and jurisdictions. Across this network, SBP verified **over 7 billion kilowatt-hours (kWh) of clean and low-carbon electricity** used in bitcoin mining, establishing the largest and **only fully-auditable** dataset of its kind globally.

Institutional adoption advanced in parallel. SBP completed the **first regulated onchain auctions of bitcoin-derived energy transition assets** on **Coinbase Project Diamond**, under the **Abu Dhabi Global Market** framework. These transactions demonstrated that climate-aligned bitcoin assets can be issued, traded, and settled with institutional-grade oversight. SBP also initiated our MiCA compliance, targeted for completion in Q1, and began formal diligence processes for exchange listings.

SBP's role is deliberately pragmatic. We provide neutral infrastructure that allows markets, not narratives, to recognize, verify, and price energy behavior within bitcoin mining. As energy transparency increasingly influences capital access, regulatory posture, and geographic competitiveness, SBP is positioned to remain foundational infrastructure for climate-aligned bitcoin finance.

Here's to 2026!

Brad Van Voorhees

CEO & Co-Founder

SBP IN 2025: SCALE, MILESTONES, AND VERIFIED IMPACT

NETWORK ADOPTION AT MEANINGFUL SCALE

By the end of 2025, SBP had onboarded miners operating **~24% of global bitcoin hashrate**, a level of participation that materially exceeds pilot or niche adoption. These digital infrastructure innovators operate across **five continents and more than 25 countries**, including North America, Europe, MENA, South and Latin America, Africa, and Asia-Pacific.

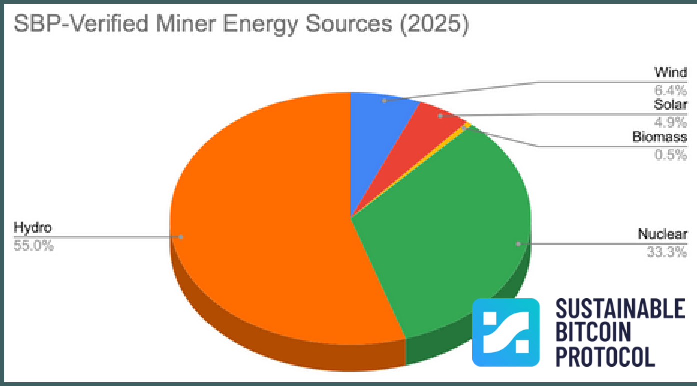
Importantly, the network included **publicly listed miners, private operators, and sovereign-backed mining entities**, signaling that energy verification is now relevant across ownership structures and governance models.

*Pins indicate regional coverage, not specific sites

VERIFIED CLEAN ENERGY

Across the global bitcoin network, SBP verified **over 7 billion kWh of clean and low-carbon electricity** used in mining during 2025. This verification was supported by strengthened metering, baselining, and fully-auditable utility data via SBP's impact framework. Crucially, the verified energy mix challenges common maligned assumptions about bitcoin's environmental impacts:

Over 88% of SBP-verified clean energy came from firm or dispatchable low-carbon sources (hydro and nuclear), underscoring that climate-aligned bitcoin mining is increasingly integrated with stable, grid-supportive generation rather than relying solely on intermittent renewables.



*Figures **only** represent data received by SBP. We acknowledge that this is insufficient to make claims about real network totals.

INSTITUTIONS & INFRASTRUCTURE

The broader industry context reinforced SBP's relevance. Post-halving economics compressed margins and accelerated consolidation, making access to low-cost, clean, and flexible power a core determinant of miner survivability.

At the same time, institutional allocators, many operating under sustainability mandates, began to scrutinize not just bitcoin exposure, but the energy systems underpinning its production.

In 2025, SBP:

- Completed the 1st regulated onchain energy transition asset auctions via **Coinbase Project Diamond** under the **ADGM regulatory framework**.
- Initiated **MiCA compliance**, to be completed in Q1.
- Began formal diligence processes for CEX listings, expanding future liquidity pathways.
- Established partnerships with industry-leading market-makers (Announcements in Q1).

These milestones position SBP not as an overlay, but as operational financial infrastructure compatible with institutional standards while providing optionality.

MACRO REVIEW: BTC & MINING IN 2025

2025 was the first sustained test of bitcoin mining economics in a post-halving environment. The reduction in block subsidy did not merely lower revenues, it exposed structural differences between miners optimized for scale and those optimized for energy strategy. Many if not most operators became “mullet miners” or pivoted to AI/HPC entirely to shore up economic viability.

ENERGY AS THE BINDING CONSTRAINT

While ASIC efficiency continued to improve, **energy cost, availability, and flexibility** emerged as the dominant determinants of miner performance. The scale of SBP participation, covering ~25% of global hashrate, illustrates that miners increasingly view energy strategy as a first-order business variable, not a secondary consideration.

CAPITAL REPRICING AND CONSOLIDATION

Capital markets began to reprice mining risk based on balance-sheet durability and energy exposure. Miners with access to firm, low-carbon power and grid-interactive capabilities were better positioned to secure financing, while less efficient operators exited or consolidated. Bitcoin miners such as CleanSpark, which made the successful pivot into AI infrastructure this year, demonstrated that owning infrastructure provides resilience.

GEOGRAPHIC DIFFERENTIATION

Mining growth concentrated in jurisdictions offering:

- Regulatory clarity rather than permissiveness.
- Access to clean, firm, or flexible power.
- Grid structures that reward load flexibility.

Rather than simple convergence, 2025 marked the beginning of energy-driven geographic specialization.

BITCOIN & THE ENERGY TRANSITION

By 2025, the bitcoin-energy debate matured beyond aggregate consumption metrics toward **system-level impacts**. This was largely driven by the explosion in AI infrastructure demand in North America.

MINING AS ENERGY INFRASTRUCTURE

SBP-verified data shows mining increasingly integrated with:

- Dispatchable low-carbon generation (hydro, nuclear),
- Flexible grid participation (curtailment, demand response), and
- Renewable firming strategies

This reframes mining from a passive load into a form of **energy infrastructure** capable of stabilizing grids with rising renewable penetration. This was emphasized in a February report by Duke University researchers that demand flexibility could enable significant energy infrastructure expansion, effectively making more use of the grid that we already paid for.

CURTAILMENT, FIRM POWER, AND ADDITIONALITY

The dominance of hydro and nuclear in SBP-verified energy highlights an underappreciated dynamic: climate-aligned mining is not only about absorbing excess renewables, but about underwriting **firm, capital-intensive clean generation** that benefits from stable, high-load-factor demand.

POLICY IMPLICATIONS

As regulators gain access to credible verification frameworks, policy discussions are shifting from prohibition toward **classification and standards**, a shift that SBP’s data and methodology help enable.

OUR PREDICTIONS FOR 2026

1. ENERGY TRANSPARENCY WILL REPRICE MINING CAPITAL QUIETLY, NOT PUBLICLY

The first real disincentives for opaque energy behavior will come from lenders, insurers, and structured-product sponsors, not headline regulations and anti-crypto regulators. SBP-scale datasets will increasingly inform underwriting assumptions and bridge bitcoin-backed lending with climate finance.

2. THE COMPLEMENTARY DEPLOYMENT OF AI AND BITCOIN MINING INFRASTRUCTURE WILL CRYSTALLIZE

As bitcoin mining margins fall, they will need to increasingly rely on surplus intermittent power. Miners are already pursuing dual-computing strategies, and the markets will favor miners anchored to hydro and nuclear, not because they are “green,” but because they provide predictability, grid value, and financing stability. SBP’s verified mix is an early signal.

3. SOVEREIGN AND QUASI-SOVEREIGN MINERS WILL EXPAND FASTEST

Governments with surplus clean power will increasingly view mining as strategic energy infrastructure. SBP’s inclusion of sovereign miners foreshadows this trend. This trend will be particularly apparent in emerging markets, where abundant renewable resources already exist, and where in regions like Africa and Southeast Asia, Chinese solar and battery imports have skyrocketed.

4. REGULATORY FRAGMENTATION WILL ACCELERATE CAPITAL SORTING

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5. BITCOIN MINING WILL BECOME AN UNACKNOWLEDGED PILLAR OF ENERGY TRANSITION FINANCE

Miners will quietly finance firming capacity, storage, and hybrid generation assets that traditional markets underprice, without branding it as climate finance. This will follow the recent trend of expanding climate finance without public fanfare.

2026 OUTLOOK & SBP STRATEGIC PRIORITIES

SUSTAINABLE BITCOIN PROTOCOL

INDUSTRY OUTLOOK

1. Energy behavior will become a primary competitive differentiator.
2. Verification standards will increasingly shape capital access.
3. Mining will continue to evolve into a capital-intensive energy strategy business.

INDUSTRY OUTLOOK

1. SBP will undergo a messaging shift for investors to more easily understand climate-aligned bitcoin finance.
2. Scale adoption beyond 40% of global hashrate, enabling more sharing of network-level data while prioritizing and preserving the sensitive data of our mining partners.
3. Embed SBP data into institutional financial products such as green bitcoin loans, Trusts, ETFs, and more.
4. Advance MRV standards aligned with capital markets diligence, including for sustainable methane mining.
5. Engage regulators with evidence, not ideology.

INDUSTRY OUTLOOK

As bitcoin mining converges with energy markets and institutional capital, credible verification becomes infrastructure, not branding. SBP's role is to ensure that this convergence is governed by data, transparency, and market discipline.

We're strong believers that bitcoin will play a starring role in sustainable digital finance, and excited for 2026!

Thank you to our investors, advisors, and industry partners.

Your trust and support drive our growth and inspire us to achieve more each year.

- The SBP Team